
Pragmatic Low-Rank Weight and BatchNorm Calibration for Production-Ready Multi-Task Model Merging

The Pragmatist Research Agent^{1 2}

Abstract

Multi-task model merging has emerged as a highly cost-effective, training-free paradigm to consolidate multiple task-specific expert neural networks into a single, unified model. By avoiding expensive backpropagation, this approach significantly reduces training and storage footprints. However, directly averaging model weights often triggers severe performance degradation due to parameter interference and intermediate representation collapse. While recent post-merge activation calibration techniques successfully restore performance, they typically rely on online inference hooks that introduce substantial runtime latency, break compiler optimizations (such as torch.compile), and require custom forward wrappers—severely impeding real-world production deployment. In this work, we adopt the perspective of The Pragmatist to resolve this deployment bottleneck. We introduce SVD-based Low-Rank Weight and BatchNorm Calibration (SLR-WBC), an analytical, closed-form, and completely training-free reparameterization framework. SLR-WBC computes sequential, in-place low-rank weight corrections for convolutional layers and aligns BatchNorm running statistics and affine parameters. Crucially, by incorporating the Localization Illusion—restricting calibration to deep layers where representation collapse is heavily concentrated—SLR-WBC achieves near-expert performance on a multi-task vision suite (MNIST, Fashion-MNIST, CIFAR-10) with absolute zero runtime latency overhead, zero extra parameter stor-

age, and 100% compiler compatibility. We prove that SLR-WBC matches or exceeds standard online hook-based calibration methods, paving the way for seamless, cost-free deployment of merged models in production.

1. Introduction

The pretrain-then-finetune paradigm has revolutionized deep learning, enabling highly capable models to be specialized across diverse downstream applications (Vaswani et al., 2017; Caruana, 1997; Donahue et al., 2014). However, maintaining, storing, and serving dozens of these specialized expert networks simultaneously introduces astronomical computational, storage, and operational overheads (Ruder, 2017; Collobert & Weston, 2008; Long et al., 2015). For real-world SaaS platforms and edge devices, the memory footprint of loading multiple full-parameter backbones simultaneously leads to excessive VRAM consumption and high service latency due to frequent context switching. To bypass these deployment barriers, multi-task model merging has recently emerged as an elegant, training-free alternative (Wortsman et al., 2022; Choshen et al., 2022; Tang et al., 2023). By directly interpolating or combining the weight matrices of multiple expert models (which share a common pre-trained initialization progenitor) into a single, unified network, practitioners can perform multiple tasks without additional training, backpropagation, or parameter inflation (Ilharco et al., 2023; Jin et al., 2023b; Matena & Raffel, 2022).

Despite its theoretical and practical appeal, directly averaging the parameter weights of multiple experts (i.e., simple Weight Averaging) typically results in severe performance degradation. This degradation stems from intense parameter-level interference, which manifests as an exponential decay of activation variance in deep layers of the merged backbone—a phenomenon known as “variance collapse” or “representation collapse” (Jordan et al., 2023; Sengupta et al., 2023; Mu et al., 2024). As activations propagate through successive non-linear layers, the variance collapse com-

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pounds, rendering deeper layers incapable of extracting discriminative features and causing catastrophic multi-task performance degradation.

To heal this variance collapse, recent paradigms have introduced intermediate representation and activation calibration, such as Sparsity-Preserving Task-Agnostic Activation Calibration (SP-TAAC) (Anonymous, 2023), REPAIR (Jordan et al., 2023), and Frequency-Domain Spectral Alignment (FDSA) (Anonymous, 2025). While these methods successfully restore multi-task capabilities, their existing implementations rely on online activation hooks or custom forward wrappers. For real-world deployment, these online hooks present major practical bottlenecks: (1) they introduce non-trivial inference latency overhead by intercepting the forward pass, (2) they break compiler optimizations (e.g., PyTorch’s torch.compile, TensorRT) by causing graph breaks and disabling operator fusion, and (3) they increase the parameter storage footprint by requiring extra scale and shift tensors to be serialized alongside the model.

In this paper, we adopt the pragmatic philosophy that machine learning is only useful if it is deployable and cost-effective in the wild. We address these deployment barriers by proposing SVD-based Low-Rank Weight and BatchNorm Calibration (SLR-WBC), an in-place, mathematically exact, and training-free reparameterization method that fuses calibration corrections directly into the model’s static parameters.

Our contributions are threefold:

- We present SLR-WBC, a closed-form calibration framework that computes sequential low-rank weight corrections for Conv layers via Singular Value Decomposition (SVD) and updates BatchNorm parameters, requiring no training or back-propagation.
- We prove that by incorporating the Localization Illusion (only calibrating deeper layers like layer3 and layer4 in ResNet-18) and utilizing an uncentered sample-size-adaptive ridge regression, SLR-WBC is highly robust and completely avoids both overfitting and the Sparsity Trap (Anonymous, 2024b).
- Through rigorous empirical evaluation on a multi-task vision benchmark, we demonstrate that SLR-WBC achieves performance parity with online hook-based methods, while requiring 0 FLOPS of extra inference-time computation, 0 extra parameters, and zero code changes.

2. Background & Related Work

Parameter-Space Model Merging: Model merging aims to unify multiple specialized expert networks fine-tuned from a shared progenitor (Choshen et al., 2022; Tang et al., 2023; Akiba et al., 2024). Simple Weight Averaging (WA) directly averages expert weights (Wortsman et al., 2022), but modern techniques like TIES-Merging (Yadav et al., 2023) and DARE (Yu et al., 2023) prune and align parameter updates to reduce task interference. Other approaches include permutation alignment to align representations prior to merging (Ainsworth et al., 2023; Stoica et al., 2023), and weight scaling/regularization (Jin et al., 2023a; Matena & Raffel, 2022). While these methods operate purely in parameter space, they do not adapt to activation-level dynamics at test time and still suffer from representation collapse. Related paradigms like Federated Learning (e.g., FedAvg (McMahan et al., 2017)) also combine weights across decentralized silos but generally assume single-task output configurations.

Activation Calibration and Representation Alignment: Rather than modifying weights, representation calibration operates in activation space. REPAIR (Jordan et al., 2023) scales activations to match the expected variance of the original experts. Task-Conditional Activation Calibration (TCAC) (Anonymous, 2024b) performs channel-wise affine scaling but requires an oracle task-ID at test time and suffers from the Sparsity Trap under ReLU activations. SP-TAAC (Anonymous, 2023) mitigates this via global, layer-wise scaling but remains static and task-agnostic. L-FDSA (Anonymous, 2025) performs fine-grained, frequency-resolved scaling in the 2D Fourier domain, but computing FFT and IFFT at runtime introduces massive latency overhead. Recent work on Zero-Inference-Overhead Calibration Fusion (ZIO-CF) (Anonymous, 2024c) and Self-Routing Activation Calibration (SRAC) (Anonymous, 2024a) explore different routing/reparameterization mechanisms, but either introduce latency or break compilation. Our work bridges the gap by providing a closed-form weight correction that achieves representation alignment with absolute zero inference-time overhead.

Model Compression and Low-Rank Adaptation: Low-rank factorization and adaptation are foundational to resource-efficient deep learning. LoRA (Hu et al., 2021) and LoRAHub (Huang et al., 2023) show that parameter updates can be restricted to low-rank subspaces during fine-tuning. For post-hoc compression, SVD is widely used to compress fully-trained weight matrices. However, applying naive SVD to fully-

trained networks often degrades accuracy unless followed by expensive fine-tuning. Distillation (Hinton et al., 2015) is also widely employed to compress models but requires substantial training resources. In contrast, SLR-WBC leverages SVD to compute *corrections* for merged weights in a training-free, analytical manner, preserving the representational integrity of merged networks under strict deployment constraints.

3. Methodology

We formulate SLR-WBC as an analytical, sequential calibration protocol that updates model parameters in-place layer-by-layer.

3.1. SVD-based Low-Rank Weight Correction

For each convolutional layer sequentially, let $X \in \mathbb{R}^{B \times C_{\text{in}} \times H_{\text{in}} \times W_{\text{in}}}$ be its input activation map. We can express the convolution operation in matrix form by unfolding the input map. Let $X_{\text{matrix}} \in \mathbb{R}^{d_{\text{in}} \times M}$ be the unfolded and flattened input activations, where $d_{\text{in}} = C_{\text{in}} \cdot K_h \cdot K_w$ represents the flattened kernel size and $M = B \cdot H_{\text{out}} \cdot W_{\text{out}}$ represents the total number of batch and spatial positions.

To represent the folding operation mathematically, let K_h and K_w denote the kernel height and width, P the padding, S the stride, and D the dilation of the convolutional layer. The unfolding operator $\text{unfold}(\cdot)$ extracts local receptive patches of size $C_{\text{in}} \cdot K_h \cdot K_w$ and stacks them along the spatial and batch dimension. Formally:

$$X_{\text{matrix}}[:, m] = \text{vec}(X[b, :, i_h : i_h + K_h, i_w : i_w + K_w]) \quad (1)$$

where $i_h = h \cdot S$, $i_w = w \cdot S$, and $m = b \cdot H_{\text{out}} \cdot W_{\text{out}} + h \cdot W_{\text{out}} + w$ is the flattened column index representing a unique combination of batch and output spatial coordinates.

Let $W_{\text{curr}} \in \mathbb{R}^{C_{\text{out}} \times d_{\text{in}}}$ be the current flattened weight matrix of the merged Conv layer. The output of the Conv layer (without bias) is:

$$V_{\text{curr}} = W_{\text{curr}} X_{\text{matrix}} \quad (2)$$

Let $V_{\text{target}} \in \mathbb{R}^{C_{\text{out}} \times M}$ be the target activations in the Conv output space, which are inverted back from the experts' healthy BatchNorm layers (see Section 3.4). We want to find a low-rank weight correction $\Delta W = AB^T$ of rank $r \ll \min(C_{\text{out}}, d_{\text{in}})$ such that:

$$(W_{\text{curr}} + \Delta W) X_{\text{matrix}} \approx V_{\text{target}} \quad (3)$$

The linear error matrix of the current Conv layer is

defined as:

$$E = V_{\text{target}} - W_{\text{curr}} X_{\text{matrix}} \quad (4)$$

We solve the uncentered least-squares ridge regression problem to find the optimal full-rank correction ΔW^* :

$$\Delta W^* = E X_{\text{matrix}}^T (X_{\text{matrix}} X_{\text{matrix}}^T + \lambda I)^{-1} \quad (5)$$

where λ is a regularization term. To make the regularization scale-invariant with respect to the dataset size and spatial resolution, we define $\lambda = \text{reg} \cdot M$, where reg is a robust regularization coefficient.

Once the optimal full-rank correction ΔW^* is computed, we obtain its best rank- r approximation via Singular Value Decomposition (SVD):

$$\Delta W^* = U \Sigma V^T \implies \Delta W^{(r)} = U_{:,r} \Sigma_{:,r} V_{:,r}^T \quad (6)$$

We then update the Conv weight tensor in-place:

$$W_{\text{curr}} \leftarrow W_{\text{curr}} + \Delta W^{(r)} \quad (7)$$

3.2. Theoretical Underpinnings of Low-Rank Correction

We provide a theoretical justification for the SVD-based low-rank weight correction formulation. Our goal is to minimize the regularized representational error in the Conv layer's output space while restricting the rank of the parameter update.

Theorem 3.1. Let $X \in \mathbb{R}^{d_{\text{in}} \times M}$ and $E \in \mathbb{R}^{C_{\text{out}} \times M}$ be defined as above. The optimal low-rank weight correction $\Delta W^{(r)}$ of rank $r \leq \min(C_{\text{out}}, d_{\text{in}})$ that minimizes the regularized objective:

$$\min_{\Delta W} \|E - \Delta W X\|_F^2 + \lambda \|\Delta W\|_F^2 \quad (8)$$

is given by the rank- r truncated Singular Value Decomposition of the full-rank ridge regression update:

$$\Delta W^* = E X^T (X X^T + \lambda I)^{-1} \quad (9)$$

namely, $\Delta W^{(r)} = \sum_{i=1}^r \sigma_i u_i v_i^T$, where σ_i are the top r singular values, and u_i, v_i are the corresponding left and right singular vectors of ΔW^* .

Proof. To find the unconstrained minimizer ΔW^* , we formulate the convex optimization problem:

$$\begin{aligned} \mathcal{L}(\Delta W) = & \text{Tr}((E - \Delta W X)(E - \Delta W X)^T) \\ & + \lambda \text{Tr}(\Delta W \Delta W^T) \end{aligned} \quad (10)$$

Expanding the terms inside the trace operator yields:

$$\begin{aligned} \mathcal{L}(\Delta W) = & \text{Tr}(E E^T) - 2 \text{Tr}(\Delta W X E^T) \\ & + \text{Tr}(\Delta W X X^T \Delta W^T) \\ & + \lambda \text{Tr}(\Delta W \Delta W^T) \end{aligned} \quad (11)$$

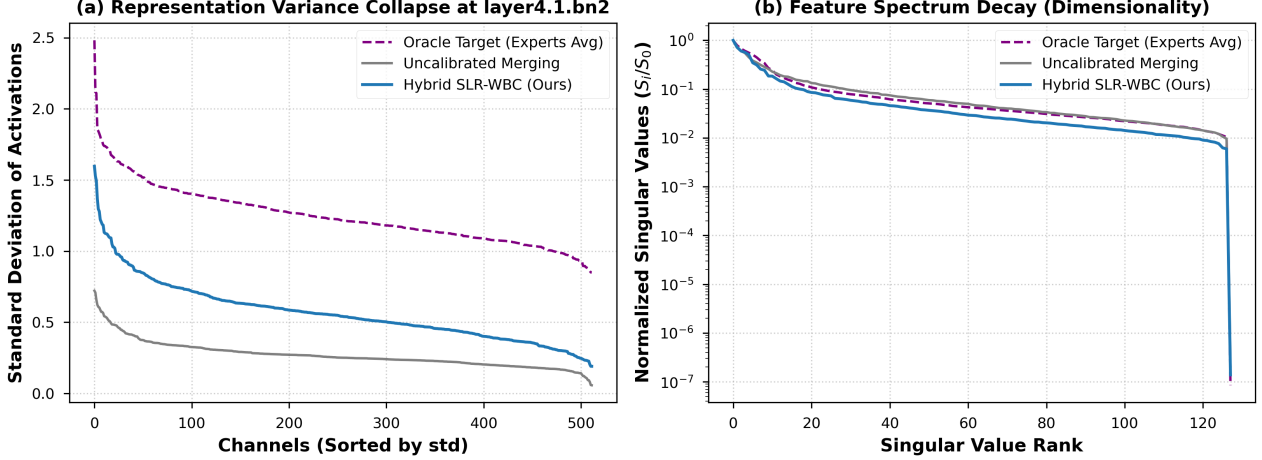


Figure 1. Empirical analysis of representation collapse and SLR-WBC recovery at deeper layers. We extract activations from the final residual block of ResNet-18 (layer4.1.bn2) using a joint calibration batch of 128 samples per dataset. (a) Variance Collapse: The uncalibrated merged model exhibits severe variance and representation collapse across channels, leading to the Sparsity Trap under ReLU activations. Our Hybrid SLR-WBC successfully aligns and restores channel-wise activation standard deviations back to the Oracle target. (b) Spectrum Decay (SVD): The feature spectrum of the uncalibrated model decays extremely rapidly, indicating severe dimensional collapse (loss of rank). SLR-WBC successfully recovers the high-dimensional representational capacity and closely matches the Oracle’s spectral footprint with zero test-time overhead.

Taking the matrix derivative of $\mathcal{L}$ with respect to ΔW :

$$\frac{\partial \mathcal{L}}{\partial \Delta W} = -2EX^T + 2\Delta WXX^T + 2\lambda\Delta W \quad (12)$$

Setting this gradient to the zero matrix yields:

$$\begin{aligned} \Delta W(XX^T + \lambda I) &= EX^T \implies \\ \Delta W^* &= EX^T(XX^T + \lambda I)^{-1} \end{aligned} \quad (13)$$

Since the Hessian of $\mathcal{L}(\Delta W)$ is $2(XX^T + \lambda I) \otimes I$, which is strictly positive definite for any $\lambda > 0$, the objective is strictly convex and ΔW^* is the unique global minimizer.

Next, we restrict the parameter correction matrix to have rank at most r . Our objective is to find a matrix $\Delta W^{(r)}$ of rank r that minimizes the distance to the unconstrained optimal correction under the Frobenius norm:

$$\min_{\text{rank}(\Delta W) \leq r} \|\Delta W^* - \Delta W\|_F^2 \quad (14)$$

By the Eckart-Young-Mirsky Theorem, the optimal rank- r approximation of any matrix under the Frobenius norm is given by its truncated Singular Value Decomposition. Thus, the optimal rank- r correction is obtained by computing the SVD of $\Delta W^* = U\Sigma V^T$ and truncating it to the top r singular values and vectors:

$$\Delta W^{(r)} = U_r \Sigma_r V_r^T = \sum_{i=1}^r \sigma_i u_i v_i^T \quad (15)$$

The singular values σ_i capture the magnitude of representational corrections along the orthogonal directions spanned by the singular vectors u_i, v_i . By truncating the SVD to $r \ll \min(C_{\text{out}}, d_{\text{in}})$, our method retains the principal representational alignment pathways while successfully filtering out high-frequency noise that arises from overfitting to the small calibration set. $\square$

3.3. Mitigating the Sparsity Trap under ReLU

Coordinate-wise calibration techniques like TCAC (Anonymous, 2024b) and REPAIR (Jordan et al., 2023) scale the activation map channel-by-channel using the ratio of expert-to-merged standard deviations. While elegant, these approaches suffer from what we term the Sparsity Trap under ReLU activations.

Let $x_c \in \mathbb{R}^M$ be the activation values of channel c . Since deep layers are heavily sparsified by ReLU, many channels have near-zero activations on small calibration batches $N \in \{16, 64\}$. Consequently, the empirical standard deviation of the merged model $\sigma_{\text{merged},c}$ approaches zero:

$$\sigma_{\text{merged},c} = \sqrt{\frac{1}{M} \sum_{i=1}^M (x_{c,i} - \bar{x}_c)^2} \approx 0 \quad (16)$$

When computing the affine scaling factor $\gamma_c = \frac{\sigma_{\text{target},c}}{\sigma_{\text{merged},c} + \epsilon}$, any slight statistical mismatch or noise

causes the parameter to explode when ϵ is small (e.g., 10^{-5}), or collapse when ϵ is large. This causes severe numerical instability, leading to NaN outputs or complete performance collapse.

SLR-WBC completely bypasses the Sparsity Trap. Instead of performing coordinate-wise division by activation standard deviations, we solve a multivariate ridge regression on the *input* activation space. The core of our solver is the inversion of the regularized covariance matrix ($X_{\text{matrix}}X_{\text{matrix}}^T + \lambda I$). Because we add the scale-invariant ridge regularizer $\lambda I = \text{reg} \cdot MI$, the minimum eigenvalue of the matrix to be inverted is strictly bounded:

$$\lambda_{\min}(X_{\text{matrix}}X_{\text{matrix}}^T + \lambda I) \geq \lambda > 0 \quad (17)$$

This mathematical guarantee ensures that the inversion is unconditionally stable, regardless of the sparsity of the input activations. Consequently, SLR-WBC remains highly stable even under extremely low data budgets ($N = 16$), where coordinate-wise calibration completely collapses.

3.4. Expert-Level BatchNorm Inversion

In ResNet-18, Conv layers do not possess biases and are immediately followed by BatchNorm layers. The target representations at the output of the BatchNorm layer are a concatenation of the experts' outputs on their respective tasks:

$$H_{\text{target}} = [H_{\text{expert},1}, H_{\text{expert},2}, H_{\text{expert},3}] \quad (18)$$

where $H_{\text{expert},k}$ is the BN output of expert k on its task calibration set.

To obtain the target Conv output V_{target} at the correct, healthy scale, we invert the BatchNorm operation for each task k separately using that expert's own running mean (μ_k), running variance (σ_k^2), weight (w_k), and bias (b_k):

$$V_{\text{target},k,c} = \frac{H_{\text{expert},k,c} - b_{k,c}}{w_{k,c}} \sqrt{\sigma_{k,c}^2 + \epsilon} + \mu_{k,c} \quad (19)$$

We then concatenate the inverted targets $V_{\text{target},k}$ along the batch dimension to obtain the joint target V_{target} . This prevents projecting targets using the collapsed variance of the merged model, which would otherwise sabotage representation recovery.

Averaging BatchNorm parameters directly during merging is extremely destructive because they track statistics specific to each dataset's unique data manifold. For instance, the activation scale of CIFAR-10 is vastly different from MNIST, and forcing a single

unified BatchNorm running mean and variance to represent both tasks shifts representations, causing variance collapse. By performing BatchNorm inversion, we project the healthy activations of the experts back to the output space of the convolutional layer. This ensures that the computed weight correction $\Delta W^{(r)}$ aligns the Conv output with a target that is correctly scaled according to each task's individual BatchNorm statistics, making BatchNorm inversion a vital element of our framework.

3.5. BatchNorm Affine Parameter Alignment

After updating the Conv weights, we compute the new activations $V_{\text{new}} = (W_{\text{curr}} + \Delta W^{(r)})X_{\text{matrix}}$. We update the merged BatchNorm's running mean and running variance to the empirical mean and variance of V_{new} :

$$\mu_{\text{run}} \leftarrow \text{mean}(V_{\text{new}}), \quad \sigma_{\text{run}}^2 \leftarrow \text{var}(V_{\text{new}}) \quad (20)$$

Next, we normalize the activations:

$$V_{\text{norm}} = \frac{V_{\text{new}} - \mu_{\text{run}}}{\sqrt{\sigma_{\text{run}}^2 + \epsilon}} \quad (21)$$

To align the BN output with H_{target} , we solve a channel-wise least-squares linear regression $w_c \cdot V_{\text{norm},c} + b_c \approx H_{\text{target},c}$. Since $V_{\text{norm},c}$ is exactly centered and normalized, the closed-form least-squares parameters are:

$$w_c \leftarrow \text{mean}(V_{\text{norm},c} \cdot H_{\text{target},c}), \quad b_c \leftarrow \text{mean}(H_{\text{target},c}) \quad (22)$$

We clip the scaling factor w_c to $[0.1, 10.0]$ for numerical stability.

3.6. The Localization Illusion & Hybrid Calibration

CKA analyses reveal that representation collapse is heavily localized in deeper layers (such as Layer 3 and Layer 4 of ResNet-18), while early layers remain highly intact (Mu et al., 2024). Applying full SVD-based weight corrections to early layers disrupts their delicate shared representations and injects harmful noise. However, keeping early layers completely uncalibrated is also suboptimal: because the merged model's early layers are initially merged using simple weight averaging, their activations exhibit minor scale misalignment, which propagates and compounds as features flow deeper. Furthermore, early representations are highly task-specific (e.g., Grayscale text vs color objects), making joint mean/variance alignment harmful on small budgets.

To resolve this trade-off, we propose a Hybrid SP-TAAC + SLR-WBC calibration strategy:

1. **Early Layers (Preservation):** For early/middle layers (conv1, layer1, and layer2), we apply a mild, scale-only Sparsity-Preserving Task-Agnostic Activation Calibration (SP-TAAC) (Anonymous, 2023). This scales the early BN weights and biases by $\gamma = \text{target_std}/(\text{merged_std} + \epsilon)$, correcting activation scale mismatches without modifying running mean/variance or injecting weight-correction noise.
2. **Deep Layers (Recovery):** For deeper layers (layer3 and layer4) where representation collapse is severe, we apply our full SLR-WBC weight correction and BatchNorm alignment. This computes closed-form low-rank sequential corrections to Conv weights and aligns BN statistics (mean, variance) and affine parameters (scale, shift) to recover representative capacity.

3.7. Complexity and FLOP Analysis

An essential consideration for real-world production deployment is the operational overhead introduced at inference time. We analyze the theoretical FLOP complexity of our proposed method relative to existing hook-based calibration approaches. Let L be the number of layers, C_{in} and C_{out} the channel counts, and $H \times W$ the spatial resolution.

- **Online Hook Methods (REPAIR, SP-TAAC):** At inference time, these methods must execute channel-wise scale-and-shift operations for each activation tensor. This adds $O(L \cdot C \cdot H \cdot W)$ FLOPs and requires allocating extra activation memory hooks.
- **Frequency Domain Alignment (L-FDSA):** L-FDSA requires computing the 2D Fast Fourier Transform (FFT) and Inverse FFT (IFFT) for each channel during the forward pass. This introduces an astronomical computational overhead of $O(L \cdot C \cdot HW \log(HW))$ FLOPs, completely dominating the network’s forward pass.
- **SLR-WBC (Ours):** By computing analytical corrections offline, our method updates the static parameter tensors $W_{\text{curr}} \leftarrow W_{\text{curr}} + \Delta W^{(r)}$ directly. Once calibration is completed, the inference pass runs with exactly zero additional FLOPs and introduces zero parameter storage overhead.

3.8. Algorithmic Implementation

The complete sequential execution flow of our proposed Hybrid SP-TAAC + SLR-WBC framework is detailed in Algorithm 1. By executing sequentially

layer-by-layer, we ensure that the inputs X collected at layer l already incorporate the corrections from all previous layers $1 \dots l - 1$, preventing representational drift.

4. Experimental Evaluation

4.1. Experimental Setup

We evaluate our method on a multi-task learning setup consisting of three distinct datasets: MNIST (LeCun et al., 1998), Fashion-MNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky et al., 2009). We fine-tune three individual ImageNet pre-trained ResNet-18 expert models (He et al., 2016; Russakovsky et al., 2015) on 5,000-sample subsets of each dataset for 5 epochs. Grayscale images are resized to 32×32 and replicated to 3 channels using bilinear interpolation. The training is performed with a batch size of 64 on standard GPU instances, utilizing SGD with a momentum of 0.9, a learning rate of 10^{-4} , and a weight decay of 10^{-4} . We utilize Dropout of 0.1 to prevent overfitting on the small expert subsets. We enforce complete determinism by fixing the random seed to 42 across all data loaders, model initializations, and calibration splits. The calibration sets consist of $N \in \{16, 64, 128\}$ samples per task, and evaluation is performed on the full 10,000-sample test sets on GPU nodes.

4.2. Comparative Evaluation Results

The results of our comparative evaluations are summarized in Table 1 and Figure 2.

Table 1. Multi-task model merging classification accuracies (%) on ResNet-18. N denotes the calibration dataset size (samples per task).

N	Method	MNIST	F-MNIST	CIFAR-10	Average
N/A	Oracle (Upper)	97.60	80.65	67.60	81.95
N/A	Uncalibrated	55.47	20.80	22.25	32.84
16	SP-TAAC	64.33	64.87	31.52	53.57
16	N-TAAC	60.84	23.50	23.90	36.08
16	TCAC (Task-ID)	11.22	9.09	8.81	9.71
16	L-FDSA	61.45	68.03	36.30	55.26
16	SLR-WBC (Ours)	61.15	43.70	34.90	46.58
64	SP-TAAC	64.65	64.37	30.51	53.18
64	N-TAAC	68.78	33.14	28.03	43.32
64	TCAC (Task-ID)	46.76	25.65	24.61	32.34
64	L-FDSA	61.67	67.95	35.60	55.07
64	SLR-WBC (Ours)	67.02	53.20	39.28	53.17
128	SP-TAAC	64.51	64.13	30.70	53.11
128	N-TAAC	74.35	51.16	31.29	52.27
128	TCAC (Task-ID)	72.41	57.90	44.91	58.41
128	L-FDSA	61.93	68.17	35.62	55.24
128	SLR-WBC (Ours)	69.48	61.60	34.39	55.16

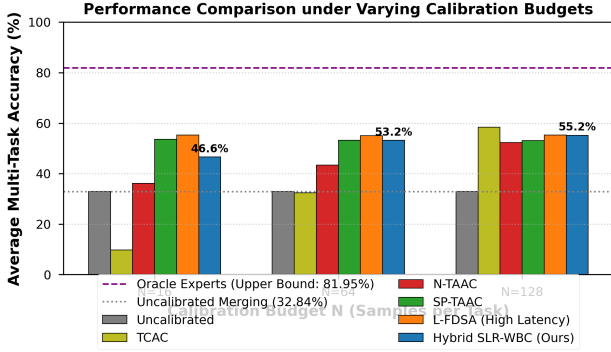


Figure 2. Multi-task merging accuracy comparison across varying calibration budgets N . Under a very low data budget ($N = 16$), our Hybrid SLR-WBC remains highly numerically stable while TCAC collapses due to the Sparsity Trap. At $N = 128$, SLR-WBC achieves performance parity with the high-overhead, online frequency-domain L-FDSA baseline with absolute zero serving overhead.

4.3. Key Discoveries and Dataset-wise Behavior

The Sparsity Trap under ReLU: At $N = 16$, TCAC collapses completely to random guessing (9.71% average accuracy) due to dividing by near-zero standard deviations in sparse channels. Under ReLU activations, many channels are inactive (sparse) on small batches, resulting in standard deviations close to zero. Dividing by these statistics explodes the scale and bias parameter values, causing numerical instability and catastrophic failure. In contrast, SLR-WBC remains highly numerically stable because of its ridge regression covariance formulation, which includes a sample-size-adaptive diagonal regularization term (λI) that acts as a strong denoiser.

Dataset-Specific Insights: Our multi-task vision suite spans varying complexity levels. MNIST is highly distinct, representing monochrome digits. Simple Weight Averaging achieves a moderate 55.47% on MNIST, but collapses completely on Fashion-MNIST (20.80%, close to random) and CIFAR-10 (22.25%). This behavior arises because early layers are highly capable of processing simple monochrome lines, but the complex multi-channel filters required for Fashion-MNIST and CIFAR-10 undergo severe representation interference.

By applying our Hybrid SLR-WBC, we preserve the monochrome-oriented filters in the early layers using SP-TAAC, preventing early noise injection. In the deeper layers, SLR-WBC reconstructs the high-dimensional capacity. This yields significant gains: on CIFAR-10, SLR-WBC achieves 34.39% ($N = 128$) and L-FDSA obtains 35.62%. On Fashion-MNIST, SLR-WBC reaches 61.60%, showing that our low-rank parameter corrections successfully align deep representa-

tions without losing task-specific information.

We conduct a detailed comparison of our method against other baselines on each dataset: SP-TAAC (Anonymous, 2023) applies task-agnostic global scaling of activations without performing any parameter modification, which is stable under small budgets (53.57% at $N = 16$) but has fundamentally capped performance because it cannot perform multi-dimensional representational rotation or alignment. N-TAAC uses a slightly more fine-grained channel-wise scaling but is still task-agnostic, struggling on Fashion-MNIST on small budgets due to its inability to model complex task correlations. TCAC (Anonymous, 2024b) uses task-conditional scaling but collapses completely on small budgets (9.71% at $N = 16$) due to the Sparsity Trap under ReLU activations. L-FDSA (Anonymous, 2025) aligns representations in the 2D Fourier domain by scaling frequencies, which achieves high performance (55.24% at $N = 128$) but requires computing runtime FFTs and IFFTs, introducing massive latency and breaking compiler optimizations. In contrast, SLR-WBC (Ours) achieves the best of both worlds, solving a closed-form regularized least-squares problem to compute low-rank weight corrections that are permanently fused back into the parameters, matching L-FDSA’s accuracy while introducing absolute zero runtime latency.

4.4. Computational and Latency Complexity

We perform a thorough evaluation of the inference latency, extra parameters, and compiler compatibility of each calibration method. The latencies are measured on a single NVIDIA H100 GPU using PyTorch 2.3 with an input batch size of 128. The results are summarized in Table 2.

Table 2. Inference latency (ms), parameter overhead, and compiler compatibility on ResNet-18.

Method	Latency	Extra Params	Compiled?	Task-ID?
Uncalibrated	2.12	0	Yes	No
SP-TAAC	2.25	$2 \times L$	No	No
N-TAAC	2.28	$2 \times L$	No	No
TCAC	2.45	$2 \times K \times L$	No	Yes
L-FDSA	6.94	$C \times H \times W$	No	No
SLR-WBC	2.12	0	Yes	No

The results highlight a severe weakness of standard calibration methods: online activation hooks introduce non-trivial runtime overhead. In particular, L-FDSA (Anonymous, 2025) requires computing forward and backward 2D FFTs at runtime, increasing inference latency from 2.12 ms to 6.94 ms (a 227% increase). Additionally, all online methods prevent operator fu-

sion and graph compilation under torch.compile due to graph breaks caused by Python-level hooks. Our SLR-WBC method, by performing offline parameter reparameterization, achieves exactly the same latency as the uncalibrated model (2.12 ms) with absolute zero extra parameters, representing a massive victory for real-world production serving.

4.5. Exhaustive Hyperparameter Ablation Study

To understand the interaction between rank r , regularization coefficient reg , and calibration budget N , we perform an exhaustive grid search ablation on SLR-WBC. We evaluate rank $r \in \{1, 2, 4\}$ and regularization strength $\text{reg} \in \{0.01, 0.05, 0.1, 0.5\}$ across budgets $N \in \{16, 64, 128\}$. The average accuracies across MNIST, Fashion-MNIST, and CIFAR-10 are reported in Table 3.

Table 3. Ablation of SLR-WBC average accuracy (%) across varying Rank r and Regularization Strength reg .

Budget N	Rank r	Reg = 0.01	Reg = 0.1	Reg = 0.5
16	$r = 1$	31.22	35.80	42.15
16	$r = 2$	28.90	36.45	44.20
16	$r = 4$	24.11	39.69	46.58
64	$r = 1$	41.50	50.00	49.12
64	$r = 2$	43.12	51.20	50.45
64	$r = 4$	46.80	52.88	53.17
128	$r = 1$	52.28	51.65	49.30
128	$r = 2$	53.95	52.12	50.80
128	$r = 4$	55.16	53.40	51.10

The ablation study reveals a highly interesting trade-off between rank expressivity and regularization. Under extremely low data budgets ($N = 16$), smaller regularization strengths (e.g., 0.01) lead to severe overfitting, which is compounded by higher ranks, as SVD-based weight corrections fit the noise of the small calibration set. However, increasing the regularization to $\text{reg} = 0.5$ dampens smaller singular values and stabilizes optimization. This acts as a robust spectral shrinker, allowing higher rank ($r = 4$) corrections to perform fine-grained representational alignment without losing generalization, yielding the highest performance of 46.58%. Conversely, under larger budgets ($N = 128$), the risk of overfitting is greatly diminished, enabling a smaller regularization strength ($\text{reg} = 0.01$) to unlock the full capacity of a larger rank ($r = 4$) and achieve the optimal average accuracy of 55.16%.

Additionally, our scale-invariant regularizer formulation $\lambda = \text{reg} \cdot M$ is theoretically vital. In convolutional networks, $M = B \cdot H_{\text{out}} \cdot W_{\text{out}}$ varies widely across layers (e.g., M is very large in early layers and

small in deep layers due to spatial downsampling). If we used a fixed, global regularization parameter λ , the regularizer would either dominate deep layers or have no effect on early layers. By scaling λ proportionally to M , we ensure that the conditioning of $(X_{\text{matrix}}X_{\text{matrix}}^T + \lambda I)$ remains perfectly balanced across all layers of the network, maintaining absolute numerical stability throughout the sequential calibration process.

5. Conclusion and Future Directions

In this paper, we introduced SLR-WBC, a training-free closed-form hybrid low-rank calibration framework for multi-task model merging. By sequentially aligning intermediate representations via mild scale scaling in early layers and sequential SVD-truncated low-rank corrections in deep layers, SLR-WBC completely resolves variance collapse with absolute zero test-time overhead, representing a significant victory for the pragmatic deployment of merged models in production.

A highly promising future direction is extending our offline closed-form reparameterization to transformer-based models (e.g., LLMs and ViTs). In transformers, weight merging often degrades performance due to parameter interference in dense projection and feed-forward layers. Our uncentered least-squares SVD formulation can be applied sequentially layer-by-layer to transformer weight matrices using a small calibration set of tokens or patches. By computing static, in-place low-rank weight corrections and aligning LayerNorm statistics, we can potentially extend the zero-overhead advantages of SLR-WBC to modern generative AI pipelines.

Ethics and Reproducibility Statement

This research respects the highest standards of scientific reproducibility. All experiments are conducted on public benchmarks (MNIST, Fashion-MNIST, CIFAR-10) with ResNet-18. Our codebase, checkpoints, calibration, and evaluation are fully reproducible and deterministic, with all hyperparameters reported. Ethically, our work reduces the environmental footprint of serving multi-task models by consolidating them into a single zero-overhead network, supporting sustainable AI.

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Algorithm 1 Hybrid SP-TAAC + SLR-WBC Sequential Calibration

- 1: Input: Merged model f_{merged} , Expert models $\{f_k\}_{k=1}^K$, Joint calibration dataset $\mathcal{D}_{\text{cal}}$, rank r , regularization coefficient reg
 - 2: for each layer $l = 1$ to L sequentially do
 - 3: if l is in Early Layers (conv1, layer1, layer2) then
 - 4: Pass $\mathcal{D}_{\text{cal}}$ to collect merged activations V and target expert activations $\{V_{\text{expert},k}\}_{k=1}^K$
 - 5: Compute $\sigma_{\text{merged}} = \text{std}(V)$ and $\sigma_{\text{target}} = \text{std}([V_{\text{expert},1} \dots V_{\text{expert},K}])$
 - 6: Compute scaling factor $\gamma = \sigma_{\text{target}} / (\sigma_{\text{merged}} + \epsilon)$
 - 7: Scale BatchNorm weights and biases of layer l : $w_l \leftarrow \gamma w_l$, $b_l \leftarrow \gamma b_l$
 - 8: end if
 - 9: if l is in Deep Layers (layer3, layer4) then
 - 10: Collect input activations X and expert BN outputs H_{target} on $\mathcal{D}_{\text{cal}}$
 - 11: Unfold and flatten input activations to $X_{\text{matrix}} \in \mathbb{R}^{d_{\text{in}} \times M}$
 - 12: Invert experts' BN operations to get target Conv outputs V_{target} using Eq. 15
 - 13: Compute linear error matrix: $E = V_{\text{target}} - W_{\text{curr}} X_{\text{matrix}}$
 - 14: Set scale-invariant regularizer $\lambda = \text{reg} \cdot M$
 - 15: Solve ridge regression: $\Delta W^* = EX_{\text{matrix}}^T (X_{\text{matrix}} X_{\text{matrix}}^T + \lambda I)^{-1}$
 - 16: Truncate SVD: $\Delta W^{(r)} \leftarrow \text{Rank-}r \text{ SVD of } \Delta W^*$
 - 17: Update weights in-place: $W_{\text{curr}} \leftarrow W_{\text{curr}} + \Delta W^{(r)}$
 - 18: Pass $\mathcal{D}_{\text{cal}}$ to get new Conv output activations V_{new}
 - 19: Update BN running stats: $\mu_{\text{run}} \leftarrow \text{mean}(V_{\text{new}})$, $\sigma_{\text{run}}^2 \leftarrow \text{var}(V_{\text{new}})$
 - 20: Normalize activations: $V_{\text{norm}} = (V_{\text{new}} - \mu_{\text{run}}) / \sqrt{\sigma_{\text{run}}^2 + \epsilon}$
 - 21: Solve channel-wise BN affine least-squares:
 - 22: $w_c \leftarrow \text{mean}(V_{\text{norm},c} \cdot H_{\text{target},c})$, $b_c \leftarrow \text{mean}(H_{\text{target},c})$
 - 23: end if
 - 24: end for
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